Assignment 3

1.Pick any one app you use daily (such as Instagram, Zomato, or PhonePe) and list 5 security measures it likely uses to protect user data.

Ans-

Step 1: Select the app

Choose Instagram as the application.

Step 2: Write the 5 security measures

No.	Security Measure	How it protects users
1	Two-Factor Authentication (2FA)	Adds an additional verification step during login.
2	HTTPS/TLS Encryption	Protects data while it travels between the device and servers.
3	Password Protection	Helps prevent unauthorized account access.
4	Login Alerts	Notifies users about suspicious or unfamiliar logins.
5	Access/Privacy Controls	Allows users to control who can access their profile and information.

Step 3: Practical verification

Open Instagram and check:mProfile → ☰ → Accounts Center → Password and security

Look for options such as:

Password

Two-factor authentication

Login activity

Login alerts

**2.Choose either the NIST Cybersecurity Framework, ISO 27001, or CIS Controls. For each of the 5 security measures you listed in the previous task, map it to the most relevant category or control in your chosen framework.

Hint: For example, 'two-factor authentication' might map to 'Access Control' in CIS or 'Protect' in NIST.**

Ans-

Instagram Security Measure	NIST CSF Category/Function	Reason

Two-Factor Authentication	Protect	Protects accounts from unauthorized access.
HTTPS/TLS Encryption	Protect	Protects information during transmission.
Password Protection	Protect	Provides authentication and access protection.
Login Alerts	Detect	Helps identify suspicious login activity.
Access/Privacy Controls	Protect	Helps control access to information and resources.

3.Given this scenario: A new food delivery app wants to prevent unauthorized access, protect payment info, and ensure customer privacy. Select 3 controls from the CIS Controls framework that would address these concerns and explain how each control helps.

Ans-

CIS Control	**Purpose**	**How it helps**
CIS Control 5 – Account Management	Manage user and administrator accounts	Prevents unnecessary or unauthorized accounts from accessing the application.
CIS Control 6 – Access Control Management	Manage access to systems and data	Ensures users get only the permissions they need.
CIS Control 3 – Data Protection	Protect sensitive data	Helps protect customer and payment-related information through appropriate data protection measures.

**4.Create a comparison table with three columns (NIST, ISO 27001, CIS Controls) and list 3 unique features or strengths of each framework based on your research.

Hint: Focus on what makes each framework different or especially useful for certain organizations.**

Ans-

NIST Cybersecurity Framework	ISO 27001	CIS Controls

Provides a flexible cybersecurity risk-management approach	Provides a formal Information Security Management System (ISMS)	Provides prioritized technical and organizational security safeguards
Organized around Identify, Protect, Detect, Respond and Recover	Supports systematic information-security risk management	Practical and action-oriented
Useful for organizations of different sizes and sectors	Can support formal certification to ISO/IEC 27001	Useful for organizations that need prioritized security actions

NIST CSF:
Best for understanding and managing cybersecurity risk in a structured but flexible way**.**

ISO 27001:
Best for organizations wanting a formal ISMS and potentially certification.

CIS Controls:
Best when an organization wants a prioritized list of practical security safeguards.

5. Use ChatGPT or another AI tool to generate a checklist of 5 security actions a small e-commerce startup should take according to the ISO 27001 framework. Copy the AI's response and briefly comment on whether you agree with each item and why.

Ans-

No.	Security Action	My Comment
1	Identify important information assets and security risks	Agree. The company needs to know what data and systems are important before protecting them.
2	Implement strong access control and user authentication	Agree. This reduces the possibility of unauthorized access to customer and company data.
3	Protect sensitive information using appropriate encryption	Agree. Encryption helps reduce the impact of data interception or unauthorized disclosure.
4	Create backup and recovery procedures	Agree. Backups help the company recover from accidental deletion, system failure or security incidents.
5	Conduct security awareness training and regularly review security controls	Agree. Employees can cause security incidents, so training and regular reviews are important.